

Network-1 Project Requirements

Project Number () Discussion Time ()		Team Grades (/10)	
Team Members			
Team Member	ID	Individual grades (/10)	Total grades (/20)
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Network-1 Project Requirements

Requirement	Details	Marks	Grade
Network Design	All networks reachable and used right devices	1	
VLSM	Correct VLSM calculations	1	
	Complete addressing table (Network, First/Last IP, Broadcast)	1	
Protocol Implementation	DHCP	1	
	DNS Server	1	
	VLAN	1	
	NAT (Static/PAT)	1	
	VPN	1	
	Static Routing	1	
	Web server	1	

Network-1 Project Documentation

Requirement	What You Must Do	How to Verify (Commands)	Required Screenshots
Network Design	Design a WAN using at least 4 routers, including PCs, switches, DNS server, and Web server	ping	Full topology + successful ping between networks

VLSM Calculation	Perform subnetting based on number of hosts needed	—	VLSM calculation table
Addressing Table	Provide Network ID, Subnet Mask, First IP, Last IP, Broadcast for each subnet	show ip interface brief	Router IP configurations
DHCP	Configure router to assign IP addresses automatically to PCs	show ip dhcp pool	DHCP + sample PC IP settings
DNS Server	Configure domain name (e.g., www.company.com) and link it to Web Server IP	—	Browser accessing website using domain
VLAN	Create VLANs and assign switch ports to each VLAN	show vlan brief	VLAN table
NAT (Static/PAT)	Configure Static NAT for servers and PAT or Dynamic NAT for users	show ip nat translations show ip nat statistics	NAT translation table
VPN	Configure VPN (Site-to-Site) between networks	show crypto isakmp sa show crypto ipsec sa	VPN status + successful ping
Static Routing	Configure static routes between all routers	show ip route	Routing table from each router
Web Server	Create and host a simple HTML page	—	Web page opened in browser
ICMP Testing	Test connectivity between all networks	ping	Successful ping + simulation mode scenarios

Final Submission Checklist

Item	Requirement
Packet Tracer File	Submit the .pkt file
Report	Include all configurations, tables, and explanations
Screenshots	Show commands and their outputs clearly
Functionality	Ensure all networks and services are working

Important Project Day Instructions

- Be on time; late arrival will result in loss of marks.
- Bring your printed documentation report and cover sheet; missing documents will result in loss of marks.
- Bring your laptop ready with the Packet Tracer file (.pkt); without it, you cannot present your project.
- All team members must understand every part of the project; anyone can be asked any question on project implementation and course concepts.
- Be polite and respectful during the discussion and the moderator has full authority to deduct marks for any inappropriate behavior.